

Task 2: Query Understanding and Decomposition

Overview

Before writing SQL, it is important to break down each natural language question into structured components. This process helps identify the correct tables, columns, filters, and joins needed to construct an accurate query.

Each question below is decomposed into five parts: Intent, Tables, Columns, Filters, and Joins.

Section A: Simple SELECT Queries (Q1–Q20)

Q1. List all products

- **Intent:** Retrieve all product records
- **Tables:** products
- **Columns:** productCode, productName, productLine, buyPrice, MSRP
- **Filters:** None
- **Joins:** None

Q2. Get all customers

- **Intent:** Retrieve all customer records
- **Tables:** customers
- **Columns:** customerNumber, customerName, city, country, phone
- **Filters:** None
- **Joins:** None

Q3. Show all orders

- **Intent:** Retrieve all order records
- **Tables:** orders
- **Columns:** orderNumber, orderDate, status, customerNumber
- **Filters:** None
- **Joins:** None

Q4. List all employees

- **Intent:** Retrieve all employee records
- **Tables:** employees
- **Columns:** employeeNumber, firstName, lastName, jobTitle, email
- **Filters:** None
- **Joins:** None

Q5. Get all offices

- **Intent:** Retrieve all office locations
- **Tables:** offices
- **Columns:** officeCode, city, country, territory, phone
- **Filters:** None
- **Joins:** None

Q6. Show all product lines

- **Intent:** Retrieve all product line categories
- **Tables:** productlines
- **Columns:** productLine, textDescription
- **Filters:** None
- **Joins:** None

Q7. List all payments

- **Intent:** Retrieve all payment transactions
- **Tables:** payments
- **Columns:** customerNumber, checkNumber, paymentDate, amount
- **Filters:** None
- **Joins:** None

Q8. Get product names and prices

- **Intent:** Retrieve product names with their pricing information
- **Tables:** products
- **Columns:** productName, buyPrice, MSRP
- **Filters:** None
- **Joins:** None

Q9. Get customer names and cities

- **Intent:** Retrieve customer names with their city locations
- **Tables:** customers
- **Columns:** customerName, city, country
- **Filters:** None
- **Joins:** None

Q10. List employee first and last names

- **Intent:** Retrieve employee names and their roles
- **Tables:** employees
- **Columns:** firstName, lastName, jobTitle

- **Filters:** None
- **Joins:** None

Q11. Get all order dates

- **Intent:** Retrieve all unique dates on which orders were placed
- **Tables:** orders
- **Columns:** orderDate
- **Filters:** None (DISTINCT applied)
- **Joins:** None

Q12. Show product vendor list

- **Intent:** Retrieve a unique list of all product vendors
- **Tables:** products
- **Columns:** productVendor
- **Filters:** None (DISTINCT applied)
- **Joins:** None

Q13. Get all product codes

- **Intent:** Retrieve all product codes with product names
- **Tables:** products
- **Columns:** productCode, productName
- **Filters:** None
- **Joins:** None

Q14. List all countries from offices

- **Intent:** Retrieve distinct countries where offices are located
- **Tables:** offices
- **Columns:** country, city, territory
- **Filters:** None (DISTINCT applied)
- **Joins:** None

Q15. Show all order statuses

- **Intent:** Retrieve all unique order status values
- **Tables:** orders
- **Columns:** status
- **Filters:** None (DISTINCT applied)
- **Joins:** None

Q16. Get all payment amounts

- **Intent:** Retrieve all payment records with their amounts
- **Tables:** payments
- **Columns:** customerNumber, checkNumber, paymentDate, amount
- **Filters:** None
- **Joins:** None

Q17. List all job titles

- **Intent:** Retrieve all unique job titles in the company
- **Tables:** employees
- **Columns:** jobTitle
- **Filters:** None (DISTINCT applied)
- **Joins:** None

Q18. Get customer phone numbers

- **Intent:** Retrieve customer contact phone numbers
- **Tables:** customers
- **Columns:** customerName, phone, country
- **Filters:** None
- **Joins:** None

Q19. Show product MSRP values

- **Intent:** Retrieve product retail prices and calculate margin
- **Tables:** products
- **Columns:** productName, MSRP, buyPrice, (MSRP - buyPrice) as margin
- **Filters:** None
- **Joins:** None

Q20. List order numbers

- **Intent:** Retrieve all order numbers with dates and statuses
 - **Tables:** orders
 - **Columns:** orderNumber, orderDate, status
 - **Filters:** None
 - **Joins:** None
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Section B: JOIN Queries (Q21–Q30)

Q21. Get orders with customer names

- **Intent:** Retrieve orders and attach the corresponding customer name
- **Tables:** orders, customers
- **Columns:** orderNumber, orderDate, status, customerName, country
- **Filters:** None
- **Joins:** INNER JOIN customers ON orders.customerNumber = customers.customerNumber

Q22. Get employees with office city

- **Intent:** Retrieve employees and show which city and country they work in
- **Tables:** employees, offices
- **Columns:** firstName, lastName, jobTitle, city, country
- **Filters:** None

- **Joins:** INNER JOIN offices ON employees.officeCode = offices.officeCode

Q23. Get payments with customer names

- **Intent:** Retrieve payment records and attach the customer name
- **Tables:** payments, customers
- **Columns:** checkNumber, paymentDate, amount, customerName, country
- **Filters:** None
- **Joins:** INNER JOIN customers ON payments.customerNumber = customers.customerNumber

Q24. Get order details with product names

- **Intent:** Retrieve order line items and replace product codes with readable names
- **Tables:** orderdetails, products
- **Columns:** orderNumber, productName, quantityOrdered, priceEach, line_total
- **Filters:** None
- **Joins:** INNER JOIN products ON orderdetails.productCode = products.productCode

Q25. Get products with product line description

- **Intent:** Retrieve products and attach the full description of their product line
- **Tables:** products, productlines
- **Columns:** productName, productLine, textDescription
- **Filters:** None
- **Joins:** INNER JOIN productlines ON products.productLine = productlines.productLine

Q26. Get customers with sales rep names

- **Intent:** Retrieve customers and show the name of their assigned sales representative
- **Tables:** customers, employees
- **Columns:** customerName, country, firstName, lastName (as sales_rep)
- **Filters:** None
- **Joins:** LEFT JOIN employees ON customers.salesRepEmployeeNumber = employees.employeeNumber (LEFT JOIN because some customers have no sales rep)

Q27. Get orders with customer city

- **Intent:** Retrieve orders and attach the city and country of the customer
- **Tables:** orders, customers
- **Columns:** orderNumber, orderDate, status, customerName, city, country
- **Filters:** None
- **Joins:** INNER JOIN customers ON orders.customerNumber = customers.customerNumber

Q28. Get employees and their manager

- **Intent:** Retrieve each employee alongside their manager's name
- **Tables:** employees (self-join)
- **Columns:** employee firstName, lastName, jobTitle, manager firstName, lastName
- **Filters:** None
- **Joins:** LEFT JOIN employees AS manager ON employees.reportsTo = manager.employeeNumber (LEFT JOIN keeps the President who has no manager)

Q29. Get orderdetails with product vendor

- **Intent:** Retrieve order line items and show the vendor of each product
- **Tables:** orderdetails, products
- **Columns:** orderNumber, productName, productVendor, quantityOrdered, priceEach
- **Filters:** None
- **Joins:** INNER JOIN products ON orderdetails.productCode = products.productCode

Q30. Get payments with customer country

- **Intent:** Retrieve payments and attach the customer's country for geographic analysis
 - **Tables:** payments, customers
 - **Columns:** checkNumber, paymentDate, amount, customerName, country
 - **Filters:** None
 - **Joins:** INNER JOIN customers ON payments.customerNumber = customers.customerNumber
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Section C: Aggregation Queries (Q31–Q50)

Q31. Count customers per country

- **Intent:** Count how many customers exist in each country
- **Tables:** customers
- **Columns:** country, COUNT(*) as customer_count
- **Filters:** None
- **Joins:** None

Q32. Total payments per customer

- **Intent:** Sum all payment amounts for each customer
- **Tables:** customers, payments
- **Columns:** customerName, COUNT(checkNumber), SUM(amount)
- **Filters:** None
- **Joins:** INNER JOIN payments ON customers.customerNumber = payments.customerNumber

Q33. Number of orders per status

- **Intent:** Count how many orders exist for each status value
- **Tables:** orders
- **Columns:** status, COUNT(*) as order_count
- **Filters:** None
- **Joins:** None

Q34. Products per product line

- **Intent:** Count how many products belong to each product line
- **Tables:** products
- **Columns:** productLine, COUNT(*) as product_count
- **Filters:** None

- **Joins:** None

Q35. Employees per office

- **Intent:** Count how many employees work at each office
- **Tables:** offices, employees
- **Columns:** city, country, COUNT(employeeNumber) as employee_count
- **Filters:** None
- **Joins:** LEFT JOIN employees ON offices.officeCode = employees.officeCode

Q36. Total stock per product vendor

- **Intent:** Sum total inventory quantity grouped by vendor
- **Tables:** products
- **Columns:** productVendor, COUNT(*), SUM(quantityInStock)
- **Filters:** None
- **Joins:** None

Q37. Average buy price per product line

- **Intent:** Calculate average buy price and MSRP for each product line
- **Tables:** products
- **Columns:** productLine, AVG(buyPrice), AVG(MSRP)
- **Filters:** None
- **Joins:** None

Q38. Orders per customer

- **Intent:** Count how many orders each customer has placed
- **Tables:** customers, orders
- **Columns:** customerName, country, COUNT(orderNumber)
- **Filters:** None
- **Joins:** LEFT JOIN orders ON customers.customerNumber = orders.customerNumber

Q39. Max MSRP per product line

- **Intent:** Find the highest retail price within each product line
- **Tables:** products
- **Columns:** productLine, MAX(MSRP), MIN(MSRP), price_range
- **Filters:** None
- **Joins:** None

Q40. Min buy price per vendor

- **Intent:** Find the lowest buy price across each vendor's products
- **Tables:** products
- **Columns:** productVendor, MIN(buyPrice), MAX(buyPrice), COUNT(*)
- **Filters:** None
- **Joins:** None

Q41. Total number of customers

- **Intent:** Count the total number of customer records
- **Tables:** customers
- **Columns:** COUNT(*) as total_customers
- **Filters:** None
- **Joins:** None

Q42. Total number of products

- **Intent:** Count the total number of products in the catalog
- **Tables:** products
- **Columns:** COUNT(*) as total_products
- **Filters:** None
- **Joins:** None

Q43. Total revenue from payments

- **Intent:** Calculate the total revenue from all payment transactions
- **Tables:** payments
- **Columns:** COUNT(*), SUM(amount), AVG(amount)
- **Filters:** None
- **Joins:** None

Q44. Average product price

- **Intent:** Calculate the average buy price and MSRP across all products
- **Tables:** products
- **Columns:** AVG(buyPrice), AVG(MSRP), AVG(MSRP - buyPrice) as avg_margin
- **Filters:** None
- **Joins:** None

Q45. Max payment amount

- **Intent:** Find the largest single payment made
- **Tables:** payments
- **Columns:** MAX(amount), MIN(amount)
- **Filters:** None
- **Joins:** None

Q46. Min payment amount

- **Intent:** Find the smallest single payment made
- **Tables:** payments
- **Columns:** MIN(amount), MAX(amount), payment_range
- **Filters:** None
- **Joins:** None

Q47. Count total orders

- **Intent:** Count the total number of orders placed
- **Tables:** orders

- **Columns:** COUNT(*), COUNT(DISTINCT customerNumber), COUNT(DISTINCT status)
- **Filters:** None
- **Joins:** None

Q48. Total quantity in stock

- **Intent:** Calculate total inventory quantity across all products
- **Tables:** products
- **Columns:** SUM(quantityInStock), AVG(quantityInStock), MIN, MAX
- **Filters:** None
- **Joins:** None

Q49. Average MSRP

- **Intent:** Calculate the average retail price across all products
- **Tables:** products
- **Columns:** AVG(MSRP), PERCENTILE_CONT(0.5) as median_msrp
- **Filters:** None
- **Joins:** None

Q50. Number of employees

- **Intent:** Count the total number of employees in the company
- **Tables:** employees
- **Columns:** COUNT(*), COUNT(DISTINCT jobTitle), COUNT(DISTINCT office-Code)
- **Filters:** None
- **Joins:** None